



# RealTech iTOLL Automation

## Technical, Practical Working, Operations & KT Guide

Windows Standalone Vehicle Access Control System - Version 1.0

| Document Item    | Value                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| Application      | RealTech iTOLL Automation                                                                                             |
| Release          | v1.0 / Windows x64                                                                                                    |
| Application Type | WPF desktop application, .NET 8, self-contained single-file EXE                                                       |
| Primary Function | Local-first RFID vehicle IN/OUT processing with cameras, barriers, signaling and asynchronous server synchronization  |
| Source Basis     | Final production source package, configuration, implementation code, release notes and server-interface documentation |

### SECURITY

This guide intentionally does not reproduce application passwords, broker credentials, camera credentials, or file-transfer passwords from the source configuration. Use the protected site configuration for those values.

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| <b>CORE RULE</b> | Gate processing is local-first. A valid local vehicle movement can continue even when Internet, MQTT, SFTP/FTP/FTPS, or the remote application is unavailable. |

## 1. System Purpose and Scope

RealTech iTOLL Automation is a Windows standalone vehicle-access application for one IN lane and one OUT lane. It combines RFID identification, vehicle-presence sensing, RTSP camera capture, boom-barrier control, traffic signal lamps, buzzer control, local SQLite data and asynchronous server integration.

### What the system performs

- Reads separate IN and OUT RFID readers.
- Receives IN/OUT vehicle sensor events from the IND/control unit.
- Matches a vehicle sensor cycle with an RFID even when either one arrives first.
- Validates registered/inactive/unregistered RFID records and configured RFID prefixes.
- Processes FREE and PAID access, including RFID balance and contractor balance modes.
- Supports OFFLINE\_ONLY and ONLINE\_OFFLINE authorization modes.
- Captures IN/OUT transaction images from RTSP streams.
- Creates local transaction records before server synchronization.
- Operates boom barriers, RED/ORANGE/GREEN lamps and lane buzzers.
- Publishes transaction JSON through MQTT and uploads images through SFTP, FTP or explicit FTPS.
- Receives server-originated recharge commands over MQTT with duplicate protection.
- Maintains Status Log and Server Log separately.

### Local-first architecture

| Sensor + RFID<br>Either order | Local Lookup<br>Vehicle / rules | Local Decision<br>Balance / trip | Capture + DB<br>Persist locally                                                                                                                     | Gate Control<br>Barrier sequence | Background Sync<br>Image + MQTT |
|-------------------------------|---------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------|
| <b>IMPORTANT</b>              |                                 |                                  | Network failure does not by itself stop normal local gate processing. Pending data stays in SQLite and is retried when server connectivity returns. |                                  |                                 |

## 2. Technology Stack and Technical Architecture

| Layer                | Implementation                    | Role                                                                   |
|----------------------|-----------------------------------|------------------------------------------------------------------------|
| Desktop UI           | WPF on .NET 8 / net8.0-windows    | Operator dashboard, protected panels, camera surfaces and logs.        |
| Hosting / DI         | Microsoft.Extensions.Hosting      | Starts and stops background workers and shared services.               |
| Local DB             | Microsoft.Data.Sqlite             | Vehicles, contractors, prices, trips, recharge audit and import audit. |
| RFID / Control I/O   | System.IO.Ports                   | Two RFID COM ports plus wired/Bluetooth control COM.                   |
| Wi-Fi control backup | TCP socket                        | Third-level control transport for IND messages and commands.           |
| Cameras              | LibVLCSharp.WPF + VideoLAN LibVLC | IN/OUT RTSP live view, reconnect/watchdog and snapshot capture.        |
| MQTT                 | Application MQTT service          | Transactions, online authorization, recharge command/ack.              |
| SFTP                 | SSH.NET                           | Encrypted image transfer.                                              |
| FTP / FTPS           | WinSCP transfer engine            | Passive FTP or explicit TLS/SSL FTPS with                              |

| Layer  | Implementation        | Role                                 |
|--------|-----------------------|--------------------------------------|
|        |                       | size verification.                   |
| Import | CsvHelper + ClosedXML | CSV/XLSX/XLSM vehicle master import. |

## Major runtime services

| Service / Class                             | Practical Function                                                                                                          |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| ApplicationCoordinatorService               | Connects HardwareGateway events to LaneProcessor.                                                                           |
| HardwareGateway                             | Opens readers/control links, decodes RFID data, parses sensor lines, sends hardware commands and performs control failover. |
| LaneProcessor                               | Core gate state machine: sensor/RFID pairing, validation, balance, trip, approval and barrier workflow.                     |
| CameraStreamService / CameraSnapshotService | Live RTSP feeds, stream recovery and transaction snapshots.                                                                 |
| ServerSyncWorker                            | Processes PendingSync trips, uploads image, publishes MQTT and marks Synced.                                                |
| VehicleApiAutoSyncWorker                    | Refreshes vehicle master data from enabled APIs on a timer.                                                                 |
| MqttRechargeSyncWorker                      | Receives recharge commands, applies idempotent balance credits and publishes acknowledgement.                               |
| InternetConnectivityWorker                  | Separately reports Windows/network Internet availability.                                                                   |
| ServerConnectivityWorker                    | Checks MQTT and selected image-upload TCP endpoint reachability.                                                            |
| StorageCleanupWorker                        | Deletes old dated image/log files using configured retention.                                                               |

### 3. Deployment, First Launch and Storage Layout

#### Standalone application package

The project publishes one self-contained Windows x64 executable named RealTechiTOLLAutomation.exe. The target machine does not require a separately installed .NET runtime. Native VLC and WinSCP dependencies are bundled and may be extracted by the .NET single-file runtime into a temporary directory while the application is running.

#### Basic deployment

1. Use Windows 10 or Windows 11 x64.
2. Copy RealTechiTOLLAutomation.exe to the approved local application folder.
3. Launch the EXE under the intended Windows operator account.
4. On first launch, the embedded default configuration is copied to the current user's Documents folder.
5. Configure hardware, cameras, server and site identity before production operation.

#### DEPLOYMENT NOTE

The reviewed source package builds a standalone EXE; it does not contain a Windows installer or automatic Windows startup registration. If automatic launch after Windows logon/boot is required, implement it as a separate site deployment control.

#### Writable runtime locations

| Location                                                                   | Purpose                                |
|----------------------------------------------------------------------------|----------------------------------------|
| %USERPROFILE%\Documents\Realtech_systems\Config\appsettings.json           | Editable runtime configuration.        |
| %USERPROFILE%\Documents\Realtech_systems\Data\vehicle-access.db            | SQLite operational database.           |
| %USERPROFILE%\Documents\Realtech_systems\Images\YYYY\MM\DD\...             | Transaction snapshots.                 |
| %USERPROFILE%\Documents\Realtech_systems\Logs>Status\YYYY\MM\DD\status.log | Hardware/lane/operator status log.     |
| %USERPROFILE%\Documents\Realtech_systems\Logs\Server\YYYY\MM\DD\server.log | Server/image/MQTT synchronization log. |

#### First-launch behavior

- Creates the Realtech\_systems folder structure.
- Migrates legacy configuration/data/log content when the new destination is empty.
- Creates or upgrades the SQLite schema.
- Starts hardware, camera, Internet, server, API sync, MQTT recharge and pending-trip synchronization services.

### 4. Application Startup Sequence

| 1                         | 2               | 3               | 4                                                                                            | 5              | 6             |
|---------------------------|-----------------|-----------------|----------------------------------------------------------------------------------------------|----------------|---------------|
| Loading settings          | Prepare storage | Create services | Upgrade DB                                                                                   | Load dashboard | Start workers |
| Startup Stage             |                 |                 | What Happens                                                                                 |                |               |
| Branding / Loading window |                 |                 | A visible startup window shows progress instead of leaving the operator with a blank screen. |                |               |
| Configuration             |                 |                 | Reads Documents\Realtech_systems\Config\appsettings.json; first run                          |                |               |

|                        |                                                                                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | extracts the embedded default.                                                                                                                                                          |
| Storage                | Ensures Data, Images, Logs and Config folders exist.                                                                                                                                    |
| Dependency setup       | Creates repositories, lane processor, hardware gateway, cameras, MQTT/image upload and UI services.                                                                                     |
| Database               | Creates/updates tables and indexes; archives older synthetic reconciliation data where applicable.                                                                                      |
| Dashboard              | Loads counters, lane cards, status panels and operator controls.                                                                                                                        |
| Hosted services        | Starts hardware I/O, server sync, vehicle API auto sync, recharge subscription, connectivity monitors and cleanup.                                                                      |
| Startup idle state     | Hardware initialization commands both lanes to CLOSED, buzzer OFF, ALL OFF, then GREEN when hardware is ready.                                                                          |
| <b>STARTUP FAILURE</b> | If a startup stage throws an exception, the loading window reports failure and the application shows the exception details before shutting down. Capture the exact message for support. |

## 5. Dashboard, Menus and Operator Functions

### Main menu

| Menu                  | Functions                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------|
| File                  | Open Capture Folder; Exit.                                                                 |
| Server Panel          | Open protected configuration panel.                                                        |
| Payment Configuration | Manage vehicle category/capacity pricing.                                                  |
| Data                  | Import registered vehicles from file; sync from API; open data folder; clear pending data. |
| Connectivity          | COM Port Settings.                                                                         |
| Admin Controls        | Open simulation / hardware test panel.                                                     |
| About                 | Application version display.                                                               |

### Dashboard information

- Separate IN and OUT flow rows and current-vehicle cards.
- Live IN and OUT camera views.
- Vehicle number, RFID, access type, balance and lane status.
- INTERNET and SERVER connectivity indicators.
- Hardware/control connection status.
- PROCESSED counter for synchronized transactions in the current local day.
- PENDING counter for transactions still waiting for server synchronization.
- Separate Status Log and Server Log panels.

### Protected functions

Admin Controls, Server Panel and Payment Configuration are protected by configured passwords. The guide does not reproduce those passwords. Passwords and server/camera credentials are stored in the runtime configuration and should be protected using Windows account permissions.

## 6. Hardware Interfaces and Control-Link Failover

### Hardware channels

| Channel             | Default Type                     | Function                                      |
|---------------------|----------------------------------|-----------------------------------------------|
| IN RFID             | Serial COM / UHF CF binary frame | Receives IN-lane EPC/RFID.                    |
| OUT RFID            | Serial COM / UHF CF binary frame | Receives OUT-lane EPC/RFID.                   |
| IND Primary Control | Wired serial COM                 | Sensor messages + boom/light/buzzer commands. |
| IND Backup 1        | Bluetooth SPP virtual COM        | Automatic control failover.                   |
| IND Backup 2        | Raw TCP / Wi-Fi terminal         | Third-level control failover.                 |

### Control failover priority

|                |                    |                |
|----------------|--------------------|----------------|
| 1<br>Wired COM | 2<br>Bluetooth COM | 3<br>Wi-Fi TCP |
|----------------|--------------------|----------------|

Only one active IND control transport is used for automation at a time. If a write fails, the active link is closed and the next available link is selected. The reconnect loop prefers returning to Wired first, then Bluetooth, then Wi-Fi. Failback is attempted during a short command-idle period so a live command sequence is not switched in the middle of a write.

RFID reader behavior

- IN and OUT RFID readers remain independent even when the control link fails over.
- UHF CF frames are buffered so split or combined Windows serial reads are decoded correctly.
- RFID publication has a short low-level repeat suppression before the lane-level duplicate window is applied.
- Close vendor RFID tools or serial test utilities before starting the application, because Windows cannot normally share an open COM port.

|                |                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTROL SAFETY | Sensor transitions are accepted only from the currently active IND control source. This prevents wired/Bluetooth/Wi-Fi backup links from creating duplicate lane events. |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 7. RFID and Vehicle Sensor Matching

A lane process starts only when both the lane vehicle sensor and a valid RFID are available. The events are order-independent.

| Arrival Order          | Behavior                                                              |
|------------------------|-----------------------------------------------------------------------|
| Sensor first           | Sensor HIGH is stored; lane waits for RFID.                           |
| RFID first             | RFID is buffered; lane waits for Sensor HIGH.                         |
| Both available         | The lane atomically starts one matched process.                       |
| RFID buffered too long | Buffered RFID expires after the SensorValiditySeconds pairing window. |
| Repeated RFID          | Suppressed by configured duplicate-read window.                       |

### Cross-lane same-RFID guard

IN and OUT have separate lane locks, so a global RFID claim prevents the same RFID from being processed simultaneously by both lanes. When one lane owns the RFID, the other lane ignores that same RFID cycle. The ownership claim is released as soon as the physical lane receives its Released/Released message, allowing the vehicle to be processed by the opposite lane without waiting for the first lane's post-release barrier delay.

### RFID validation order

| Normalize<br>Uppercase / trim | Prefix<br>Allowed rule | Cross-lane<br>Global claim | Duplicate<br>Time window                                                                                                               | Vehicle lookup<br>Registered? | Active<br>Enabled? |
|-------------------------------|------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------|
| UNREGISTERED                  |                        |                            | An unregistered RFID causes the lane to send a RED command with the display-friendly text 'NOT Registered'. The barrier does not open. |                               |                    |

## 8. IN Lane - Practical Working

| Step | IN Operation                                                                                                                                     |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | IN sensor and RFID are matched for one physical lane cycle.                                                                                      |
| 2    | RFID is normalized/validated and the vehicle master is loaded.                                                                                   |
| 3    | The control unit receives IN RED plus the vehicle number for a registered RFID.                                                                  |
| 4    | Inactive/unregistered data is rejected; registered active data continues.                                                                        |
| 5    | The application selects RFID or Contractor balance according to Balance Support Mode.                                                            |
| 6    | Vehicle category/capacity price is resolved for PAID access; FREE debit is zero.                                                                 |
| 7    | If ONLINE_OFFLINE is enabled, an MQTT authorization request is attempted. Server REJECT is final; timeout/unavailable falls back to local rules. |
| 8    | The application checks whether a previous unmatched IN trip still exists. If yes, Missing OUT approval rules are applied.                        |

| Step | IN Operation                                                                                                                                               |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9    | After the process-wait timer, the current IN image is captured.                                                                                            |
| 10   | Trip creation and balance update are committed locally. Where applicable, the previous active IN is exceptionally closed in the same database transaction. |
| 11   | The UI shows approval and the lane runs the approved barrier sequence.                                                                                     |
| 12   | The new trip remains PendingSync until background synchronization succeeds.                                                                                |

### PAID IN local fallback rule

|                     |                                                                                                                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BALANCE RULE</b> | When online authorization is unavailable, a PAID record with a balance already below zero is rejected. A non-negative balance may receive one entry even when the debit makes the new balance negative. The following PAID entry is then rejected until balance is restored. |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### BOTH balance mode

With Balance Support Mode = BOTH, a vehicle that has a ContractorCode uses the shared contractor balance; a vehicle without a contractor assignment uses its RFID/vehicle balance. One trip never debits both balances.

## 9. OUT Lane - Practical Working

| Step                                                                                                                                                                                                                                         | OUT Operation                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                                                                                                                                                                                                                            | OUT sensor and RFID are matched.                                                                                                            |
| 2                                                                                                                                                                                                                                            | Vehicle registration and active status are checked.                                                                                         |
| 3                                                                                                                                                                                                                                            | The selected local balance source is loaded for authorization context, but OUT does not debit an amount.                                    |
| 4                                                                                                                                                                                                                                            | If ONLINE_OFFLINE is enabled, the server may approve/reject the OUT. Server REJECT is final; unavailable/timeout falls back to local rules. |
| 5                                                                                                                                                                                                                                            | The application searches for an unmatched active IN trip for the same RFID.                                                                 |
| 6                                                                                                                                                                                                                                            | If an active IN exists, the OUT trip links to that real entry transaction.                                                                  |
| 7                                                                                                                                                                                                                                            | If no active IN exists, Missing IN approval rules apply.                                                                                    |
| 8                                                                                                                                                                                                                                            | After the process-wait timer, the OUT image is captured.                                                                                    |
| 9                                                                                                                                                                                                                                            | The physical OUT transaction is saved locally with debitAmount = 0.                                                                         |
| 10                                                                                                                                                                                                                                           | The approved barrier sequence is executed and the trip is queued for server synchronization.                                                |
| <b>NO SYNTHETIC TRIP</b>                                                                                                                                                                                                                     |                                                                                                                                             |
| For Missing IN or Missing OUT approval, only the current physical RFID-reader trip is stored/uploaded/counted. The missing direction is sent as a notification in the current trip; the application does not create a synthetic transaction. |                                                                                                                                             |

## 10. Barrier, Signal and Buzzer Operation

### Approved movement sequence

|                                      |                         |                            |                                  |
|--------------------------------------|-------------------------|----------------------------|----------------------------------|
| <b>ALL OFF</b><br>Clear lane outputs | <b>ORG</b><br>Orange ON | <b>Buzzer</b><br>Buzzer ON | <b>OPEN</b><br>Boom barrier open |
|--------------------------------------|-------------------------|----------------------------|----------------------------------|

The release cycle is armed before the OPEN command. This allows a Released/Released message received during the open transition to be remembered without starting the close timer before the OPEN command has been issued.

### Vehicle release completion

|                                          |                                    |                                |                                        |                                 |                        |
|------------------------------------------|------------------------------------|--------------------------------|----------------------------------------|---------------------------------|------------------------|
| <b>Release event</b><br>IN/OUT Realeased | <b>Delay</b><br>Configured seconds | <b>CLOSE</b><br>Barrier closes | <b>Buzzer OFF</b><br>Dedicated command | <b>ALL OFF</b><br>Clear outputs | <b>GRN</b><br>Green ON |
|------------------------------------------|------------------------------------|--------------------------------|----------------------------------------|---------------------------------|------------------------|

- Raw LOW/CLEAR/OFF/NOT DETECTED text is intentionally not treated as lane completion.
- Both the configured spelling and common Released/Realeased aliases are accepted.
- Duplicate release messages do not execute the completion sequence twice.
- A best-effort uncanceled Buzzer OFF is issued in the sequence cleanup so exceptions/cancellation do not leave the buzzer latched ON.

|                   |                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>IDLE STATE</b> | On hardware initialization and rejection cleanup, the lane is driven to CLOSED barrier, Buzzer OFF, ALL OFF, then GREEN. |
|-------------------|--------------------------------------------------------------------------------------------------------------------------|

## 11. Payment, Balance and Online/Offline Authorization

### Processing modes

| Mode           | Behavior                                                                                                                                    |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| OFFLINE_ONLY   | No online trip authorization wait. Uses local cached vehicle/contractor/category data and local balance rules.                              |
| ONLINE_OFFLINE | Sends an MQTT authorization request. A valid server REJECT is authoritative. Timeout/unavailable connection falls back to local processing. |

### Balance support modes

| Mode       | Selection                                                                                     |
|------------|-----------------------------------------------------------------------------------------------|
| RFID       | Use vehicle/RFID balance for all vehicles.                                                    |
| CONTRACTOR | Use contractor balance. Vehicles without a valid active contractor cache record are rejected. |
| BOTH       | Contractor-linked vehicles use contractor balance; unlinked vehicles use RFID balance.        |

### Vehicle category pricing

The PAID IN debit is resolved from the cached/API category debit amount when present; otherwise the configured Payment category price is used. FREE access always uses a zero debit.

### Online authorization request

The request carries a unique requestId, site/device identity, direction, vehicle/RFID, category, debit amount, contractor information, selected balance source and current local balance. The response must return the same requestId. A server-provided debit amount may override the local debit for that movement.

| AUTHORITY RULE | Server approval/rejection is used only when a valid response is available. Connection failure or timeout is treated as 'server unavailable', not as a rejection. |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 12. Exceptional Approval and Auto Approval

| Case        | Normal Requirement                    | Approval Result                                                                                       |
|-------------|---------------------------------------|-------------------------------------------------------------------------------------------------------|
| Missing OUT | New IN finds an earlier unmatched IN. | Earlier active IN is closed internally; current IN proceeds; server is notified that OUT was missing. |
| Missing IN  | OUT has no active IN.                 | Current OUT proceeds; no synthetic IN is created; server is notified that IN was missing.             |

### Manual exceptional approval

- Popup opens only when the corresponding IN/OUT exceptional approval option is enabled.
- Approver name, role and mobile number are captured.
- Name input is restricted to alphabetic characters/spaces; mobile is digit-only and length-validated.
- Cancel/close rejects the current movement and keeps the barrier closed.

## Auto approval

IN and OUT auto approval can be enabled separately. When enabled, the popup is skipped and the approval is recorded as automatic with system-generated approver details.

|              |                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AUDIT</b> | Approval mode, reason, approver fields and approval timestamp are stored in the local trip record and included in the server payload for the current physical transaction. |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 13. Camera Streaming and Transaction Images

| Feature            | Practical Behavior                                                                                           |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| Live feeds         | Separate IN and OUT RTSP streams rendered with LibVLCSharp.                                                  |
| Transport          | RTSP over TCP is enabled by default.                                                                         |
| Decode             | Software decoding is the default because it is more reliable with some Windows graphics drivers.             |
| Network cache      | Default 1200 ms.                                                                                             |
| Stream watchdog    | Playback timestamp is monitored; a frozen stream is restarted (default 12 seconds).                          |
| Reconnect          | Failed streams retry using the configured reconnect interval.                                                |
| Snapshot           | Current lane snapshot is captured for the transaction after processing wait.                                 |
| Unavailable stream | Capture status records the failure; the implementation can persist a placeholder rather than block the gate. |
| Filename           | Configured prefix + timestamp + lane + transaction ID.                                                       |

### Image storage pattern

```
<Documents>\Realtech_systems\Images\YYYY\MM\DD\<lane or dated structure>\...
```

The exact image path is stored on the trip. After successful remote upload, the remote path and upload timestamp are also stored so an MQTT retry can reuse the already-uploaded image.

## 14. SQLite Database and Local Data Model

The local database uses SQLite with WAL journaling, NORMAL synchronous mode and foreign keys enabled.

| Table                   | Purpose                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| vehicles                | RFID master, vehicle number, category/capacity, FREE/PAID type, balance, active state, contractor link and optional category debit cache. |
| contractors             | Contractor/user code, name, shared balance and active state.                                                                              |
| vehicle_category_prices | Locally cached category debit values.                                                                                                     |
| trips                   | IN/OUT transaction history, balances, images, sync state, retry state, approval data and online-authorization metadata.                   |
| rfid_recharges          | Idempotent server recharge audit including raw payload and local/server balance details.                                                  |

| Table        | Purpose                            |
|--------------|------------------------------------|
| import_audit | Manual/API import summary records. |

### Transaction states

- A newly created physical trip is stored locally as PendingSync.
- Image upload state is tracked separately from the final transaction synchronization state.
- After image handling and MQTT QoS acknowledgement succeed, the trip is marked Synced.
- Failed sync attempts store error, attempt count, last-attempt time and next retry time.

|                  |                                                                                                                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ATOMICITY</b> | IN trip creation and local balance update use SQLite transactions so the trip and debit are committed together. The negative-balance guard is enforced inside the database update path during local fallback. |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 15. Vehicle Master Import and API Auto Sync

### Manual import

The application accepts CSV plus Excel XLSX/XLSM formats. It recognizes multiple column aliases and can detect Excel content accidentally saved with a .csv extension.

| Data                | Typical Accepted Names                                              |
|---------------------|---------------------------------------------------------------------|
| RFID                | rf_id, rfid, rfid_number, tag_number                                |
| Vehicle             | vehicle_no, vehicle_number, vehiclno                                |
| Category / Capacity | vehicle_category, category, vehicle_capacity, capacity              |
| Access Type         | rfid_type, access_type, type                                        |
| Balance             | balance, wallet_balance, amount                                     |
| Active Status       | rf_status, active, is_active, status                                |
| Contractor          | contractor_id / contractor_code and related contractor/user aliases |
| Category Debit      | debit_amount / category_debit_amount / trip_amount / price          |

- RFIDs are normalized to uppercase with embedded whitespace removed.
- Import prefix validation is separate from live lane prefix validation.
- Invalid/incomplete rows can be skipped instead of failing the full import.
- Equivalent repeated RFID rows are consolidated; conflicting assignments are reported for review.
- UpdateExistingRecords allows later imports/API sync to refresh master values.

### Vehicle CSV API synchronization

- Multiple API sources can be configured in Server Panel > Storage & Import.
- Each enabled source receives POST JSON containing its configured username.
- Sources have Enabled, Priority, Endpoint, Username and Request Timeout settings.
- Lower-priority sources are imported first; the highest-priority source wins conflicting values.
- Auto sync is enabled by configuration and runs at the configured cycle interval (default 30 seconds).
- Unchanged rows are not rewritten unnecessarily.

## 16. Server Synchronization - MQTT + Image Upload

Complete MQTT topic and JSON packet samples are provided in Appendix D.

## Pending trip synchronization order

| 1                      | 2            | 3          | 4                                                                                                       | 5              | 6           |
|------------------------|--------------|------------|---------------------------------------------------------------------------------------------------------|----------------|-------------|
| Read eligible pending  | Upload image | Build JSON | Publish MQTT                                                                                            | Receive PUBACK | Mark Synced |
| Behavior               |              |            | Technical Detail                                                                                        |                |             |
| Cycle                  |              |            | Configured interval; default 5 seconds.                                                                 |                |             |
| Batch                  |              |            | Configured maximum; default 25 transactions.                                                            |                |             |
| Retry                  |              |            | Exponential retry delay from InitialRetryDelaySeconds up to MaxRetryDelaySeconds.                       |                |             |
| Image already uploaded |              |            | Stored remote_image_path is reused on later MQTT retry.                                                 |                |             |
| Missing local image    |              |            | Logged, but transaction JSON can still be published.                                                    |                |             |
| MQTT success           |              |            | QoS 1 PUBACK is treated as synchronization success.                                                     |                |             |
| Server application ACK |              |            | Not currently required for gate Device_Response; PUBACK is the current final gate-sync acknowledgement. |                |             |

## Image upload protocols

| Mode          | Implementation / Notes                                                              |
|---------------|-------------------------------------------------------------------------------------|
| SFTP          | SSH.NET authenticated SFTP; encrypted transport.                                    |
| FTP           | WinSCP transfer engine in passive FTP mode; unencrypted unless TLS enabled.         |
| FTPS Explicit | FTP + explicit TLS/SSL, normally port 21. Supports certificate fingerprint pinning. |
| Disabled      | No image transfer; MQTT transaction can still synchronize.                          |

## FTPS reliability controls

- WinSCP transfer engine is used for FTP/FTPS to improve production TLS/data-channel interoperability.
- Resume is disabled and retries overwrite the destination from byte zero.
- Remote file size is read back and compared with local file size before success is accepted.
- Normal Windows certificate validation is preferred. SHA-256 fingerprint pinning is available for known legacy/self-signed servers.
- Allow Invalid FTPS Certificate is an unsafe emergency fallback and should not be a normal setting.

## 17. RFID Recharge Synchronization

A persistent MQTT subscriber can receive server-originated PAID RFID recharge commands.

| Receive<br>Recharge MQTT | Validate<br>JSON + IDs | Check ID<br>rechargeId | Apply<br>Atomic balance                                                                                                                                                                    | Audit<br>rfid_recharges | ACK<br>Publish result |
|--------------------------|------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------|
| Rule                     |                        |                        | Behavior                                                                                                                                                                                   |                         |                       |
| Idempotency              |                        |                        | rechargeId is the unique key. Redelivery of the same recharge does not credit the balance again.                                                                                           |                         |                       |
| Amount                   |                        |                        | rechargeAmount must be positive.                                                                                                                                                           |                         |                       |
| Vehicle/RFID             |                        |                        | Configured site/device/RFID/vehicle data is validated when present.                                                                                                                        |                         |                       |
| FREE RFID                |                        |                        | Recharge is rejected where business rules require PAID balance.                                                                                                                            |                         |                       |
| Balance consistency      |                        |                        | When previousBalance and newBalance are supplied, arithmetic consistency is validated.                                                                                                     |                         |                       |
| Application ACK          |                        |                        | Result includes Applied, Duplicate or Rejected information and balance values.                                                                                                             |                         |                       |
| Broker delivery          |                        |                        | Recharge uses QoS 1 behavior; persistent-session handling supports safe redelivery.                                                                                                        |                         |                       |
| RETRY RULE               |                        |                        | The server should retry the same logical recharge with the same rechargeId until it receives the matching application acknowledgement. A new rechargeId represents a new credit operation. |                         |                       |

## 18. Connectivity, Logs, Retention and Diagnostics

### Internet vs Server status

| Indicator | What It Means                                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INTERNET  | Checks Windows network availability and configured public HTTP endpoints. If probes are blocked but an interface exists, it can show network connected rather than incorrectly showing Offline.      |
| SERVER    | Performs lightweight TCP reachability checks for MQTT and the enabled image-upload port. It reports connected, partially connected or offline. This is reachability, not a full authentication test. |

### Logs

| Log        | Contains                                                                                                           |
|------------|--------------------------------------------------------------------------------------------------------------------|
| Status Log | RFID/sensor events, lane processing, hardware commands, COM/control failover, approvals and operational errors.    |
| Server Log | Sync cycles, transaction IDs, image upload, MQTT publish/ack, retry timing, server failures and recharge activity. |

### Automatic storage cleanup

- Enabled by default in the source configuration.
- Runs at application startup, when requested after Save & Apply, and every 24 hours.



- Deletes only files inside dated YYYY/MM/DD image/log folders older than configured retention.
- Database is not deleted by storage cleanup.
- Default image retention: 30 days; default log retention: 30 days.

## 19. Configuration Reference

### Server Panel tabs

| Tab                 | Main Settings                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| Device & Processing | Site/device/lane identity, timers, RFID prefixes, processing mode, balance mode, authorization timeout.                 |
| Hardware            | Simulation mode, line terminator, reconnect, UI indicators, IN/OUT reader and control serial settings, sensor messages. |
| Cameras             | Storage, prefix/timestamp, IN/OUT RTSP settings, caching, reconnect, watchdog, TCP and decoding.                        |
| Server              | Sync intervals/retry, MQTT, recharge/authorization topics, SFTP/FTP/FTPS image upload.                                  |
| Storage & Import    | DB/log paths, retention, Internet checks, import behavior, multiple Vehicle CSV APIs.                                   |
| Approval & Security | IN/OUT exceptional approval, auto approval and protected panel passwords.                                               |

### Save & Apply behavior

The Server Panel validates values, updates in-memory options, reapplies hardware settings, reloads camera configuration and requests storage cleanup. A database path change explicitly requires application restart. For MQTT/recharge/server endpoint changes, restart the application after saving so long-running hosted connections are recreated cleanly.

### Important default processing values

| Setting                      | Source Default                                                                                                                                                                                     |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ProcessWaitSeconds           | 3                                                                                                                                                                                                  |
| BarrierAndGreenDelaySeconds  | 5                                                                                                                                                                                                  |
| DuplicateReadSeconds         | 10                                                                                                                                                                                                 |
| SensorValiditySeconds        | 30                                                                                                                                                                                                 |
| VehicleDisplayResetSeconds   | 20                                                                                                                                                                                                 |
| Allowed RFID prefix          | E2                                                                                                                                                                                                 |
| Processing mode              | OFFLINE_ONLY                                                                                                                                                                                       |
| Balance support mode         | BOTH                                                                                                                                                                                               |
| Online authorization timeout | 8 seconds                                                                                                                                                                                          |
| <b>CONFIG CONTROL</b>        | Treat appsettings.json as a controlled file. It contains operational endpoints and credentials. Back it up before major changes and do not distribute an unredacted copy in general documentation. |

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## 20. Daily Operation SOP

---

### Start-of-shift / startup checks

- [ ] Windows PC has stable power and correct date/time.
- [ ] IN/OUT RFID readers and IND/control hardware are powered.
- [ ] Vendor COM test tools are closed.
- [ ] Application starts without startup error.
- [ ] Hardware status is Ready or identifies the expected failover transport.
- [ ] IN and OUT camera streams are live.
- [ ] INTERNET and SERVER indicators are understood; server outage alone does not stop local gate operation.
- [ ] Pending counter is reviewed.
- [ ] IN and OUT barriers are physically closed and lane signals are in expected idle state.

### Normal operator monitoring

- Watch the lane card for vehicle/RFID/status consistency.
- Confirm an approved movement opens the correct barrier only.
- Use Status Log for lane/hardware issues and Server Log for synchronization issues.
- Do not repeatedly scan the same tag during a live sensor cycle; duplicate/cross-lane guards may intentionally ignore it.
- Use protected Admin Controls only for authorized hardware testing or recovery.

### End-of-shift / controlled exit

- Review Pending count and server status.
- Record unresolved transaction IDs/errors if pending data cannot synchronize.
- Use File > Exit for a controlled shutdown so hosted services receive the application stop signal.

## 21. Troubleshooting and Recovery

| Symptom                           | Meaning / First Check                                                                       | Action                                                                                                                                                                                  |
|-----------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application does not open         | Capture startup error. Verify writable Documents profile and configuration JSON.            | Correct config/path issue; relaunch.                                                                                                                                                    |
| IN/OUT RFID not reading           | Check COM port, baud/read mode, cable/power; close vendor utility using the port.           | Connectivity > COM Port Settings; Save & Connect.                                                                                                                                       |
| Sensor not detected               | Check active control transport and exact sensor line. Only active source drives automation. | Review Status Log/control link and configured Sensor Messages.                                                                                                                          |
| RFID waits for sensor             | Expected when RFID arrived first.                                                           | Trigger/verify lane sensor within SensorValiditySeconds.                                                                                                                                |
| Sensor waits for RFID             | Expected when sensor arrived first.                                                         | Verify reader/antenna and tag.                                                                                                                                                          |
| Same RFID ignored on other lane   | Cross-lane guard is active while first lane sensor remains active.                          | Wait for physical Released/Released event.                                                                                                                                              |
| Unregistered RFID                 | No vehicle master match; RED NOT Registered is sent.                                        | Import/correct vehicle master record.                                                                                                                                                   |
| Inactive RFID                     | Master record is inactive.                                                                  | Correct master data if authorized.                                                                                                                                                      |
| PAID IN rejected negative balance | Offline fallback sees balance already < 0.                                                  | Recharge or correct authorized balance source.                                                                                                                                          |
| Missing IN/OUT approval popup     | Trip history is incomplete and manual approval is enabled.                                  | Approve with valid details or cancel/reject.                                                                                                                                            |
| Barrier does not move             | Check IND transport and command log; wired may have failed over.                            | Confirm Wired -> Bluetooth -> Wi-Fi link and control hardware.                                                                                                                          |
| Buzzer remains ON                 | Dedicated Buzzer OFF should be sent during completion/cleanup.                              | Use protected Admin Control Buzzer OFF; inspect control logs.                                                                                                                           |
| Camera frozen                     | Stream watchdog should restart it; check RTSP reachability and decode settings.             | Review camera config; test software decoding/TCP.                                                                                                                                       |
| SERVER partially connected        | One of MQTT or image-upload TCP endpoints is unavailable.                                   | Use Server Log and test the specific endpoint.                                                                                                                                          |
| Pending count increasing          | Server/image/MQTT sync is failing or in retry backoff.                                      | Inspect Server Log transaction stage and next retry.                                                                                                                                    |
| FTPS certificate error            | Certificate does not pass normal validation.                                                | Verify in WinSCP; use correct SHA-256 fingerprint pinning.                                                                                                                              |
| FTP/FTPS partial transfer         | WinSCP engine verifies remote size; transaction stays pending on mismatch.                  | Review Server Log and server/network path.                                                                                                                                              |
| Vehicle API sync fails            | Endpoint/username/timeout/source priority issue.                                            | Check Status Log and Storage & Import source settings.                                                                                                                                  |
| Old images/logs not deleted       | Cleanup disabled, retention not reached, or files not in dated folders.                     | Check Storage settings and folder structure.                                                                                                                                            |
| <b>RECOVERY ORDER</b>             |                                                                                             | Prefer the smallest corrective action: fix configuration/connection -> reapply hardware/camera settings -> restart application -> reboot Windows only when OS/driver state requires it. |

## 22. Backup, Restore and Change Control

### Recommended backup set

- Config: Documents\Realtech\_systems\Config\appsettings.json

- Database: Documents\Realtech\_systems\Data\vehicle-access.db
- Images: Documents\Realtech\_systems\Images (when operational evidence must be preserved)
- Logs: Documents\Realtech\_systems\Logs (for incident/support evidence)
- The deployed RealTechiTOLLAutomation.exe and its version/hash.

### Safe manual database backup

6. Close the application before copying the SQLite database for a simple file-level backup.
7. Copy the database and configuration to a dated, access-controlled location.
8. Record application version and site/device ID with the backup.
9. For restore, stop the application, preserve the current files, restore the matched configuration/database set, then launch and validate counters/hardware.

#### NOTE

The reviewed application contains storage-retention cleanup, but it does not implement an automatic database backup/restore subsystem. Database backup should therefore be part of site operations.

## 23. Build and Release Procedure

### Build environment

- Windows build computer.
- Visual Studio 2022 with .NET desktop development workload, or .NET 8 SDK.
- Internet/package access for NuGet restore.

### One-click build

1. Extract the source package.
2. Double-click build-standalone-exe.bat
3. Wait for restore and publish.
4. Output:  
Publish\StandaloneWinX64\RealTechiTOLLAutomation.exe

### Publish characteristics

| Property                  | Value                                  |
|---------------------------|----------------------------------------|
| Runtime                   | win-x64                                |
| Self-contained            | Yes                                    |
| Single file               | Yes                                    |
| Native/content extraction | Enabled for bundled VLC/WinSCP content |
| Trimming                  | Disabled                               |
| ReadyToRun                | Disabled                               |
| Debug symbols in Release  | Disabled                               |

The publish script validates that the publish folder contains only the EXE. If additional files remain, it fails rather than reporting a false single-file package.

### Release verification

- Confirm About displays v1.0.

- Confirm executable icon/branding is RealTech iTOLL Automation.
- Launch on a clean Windows x64 test machine without a separately installed .NET runtime.
- Run simulation test before field installation.
- Perform real COM/camera/server tests on the intended site hardware.

## 24. KT - Knowledge Transfer Plan

The KT should make the receiving engineer capable of operating the system, diagnosing the correct layer, changing approved settings and explaining the transaction path without reading the source code.

### Recommended KT agenda (60-90 minutes)

| Module          | Practical Demonstration                                             | Target Understanding                          |
|-----------------|---------------------------------------------------------------------|-----------------------------------------------|
| Architecture    | Trace Sensor/RFID -> LaneProcessor -> DB -> barrier -> server sync. | Local-first and asynchronous server design.   |
| Startup         | Launch application and identify loading stages.                     | Where startup can fail.                       |
| Hardware        | Show IN/OUT readers and active control link.                        | Wired -> Bluetooth -> Wi-Fi failover.         |
| Lane processing | Run one IN and one OUT in simulation/test mode.                     | Sensor/RFID matching and physical trip rules. |
| Balance         | Demonstrate FREE, PAID and selected balance mode.                   | Debit at IN; no debit at OUT.                 |
| Exceptions      | Demonstrate Missing IN/OUT approval in test environment.            | No synthetic missing transaction.             |
| Cameras         | Show live feeds and saved image folder.                             | Capture and recovery behavior.                |
| Server          | Disconnect/reconnect test network in a safe test environment.       | PendingSync, retry and eventual sync.         |
| Data            | Import a test master/API source.                                    | Normalization, priorities and audit.          |
| Logs            | Locate transaction in Status and Server logs.                       | Use correct log for the failure layer.        |
| Configuration   | Open protected Server Panel and Hardware Settings.                  | What can apply live and what needs restart.   |
| Recovery        | Walk through troubleshooting matrix.                                | Smallest-layer recovery first.                |

### Hands-on KT exercises

- [ ] Locate Config, Data, Images and Logs folders.
- [ ] Identify current application version.
- [ ] Run a simulation-mode IN trip with a registered RFID.
- [ ] Run the matching OUT trip.
- [ ] Explain why a second IN without OUT becomes a Missing OUT condition.
- [ ] Explain when the one-negative-trip PAID rule applies.
- [ ] Identify the active IND control transport.
- [ ] Locate a transaction image and the corresponding Status/Server log lines.
- [ ] Explain Pending vs Processed counters.
- [ ] Explain image-upload-first then MQTT synchronization.
- [ ] Explain why MQTT PUBACK is not the same as an application-level server ACK.
- [ ] Explain rechargeId duplicate protection.
- [ ] Perform a controlled application restart.

## KT validation questions

| Question                                                  | Expected Technical Answer                                                                                                                                         |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Can the gate run if MQTT is down?                         | Yes. Local processing continues; trips remain PendingSync and retry later.                                                                                        |
| When does a lane process start?                           | When sensor HIGH and a valid RFID are both present for that lane within the pairing window.                                                                       |
| Can the same RFID process in IN and OUT at the same time? | No. A global cross-lane RFID claim prevents concurrent processing.                                                                                                |
| When is the RFID claim released?                          | On the physical lane Released/Released event, not after the entire barrier delay.                                                                                 |
| What is the control failover order?                       | Wired COM -> Bluetooth COM -> Wi-Fi TCP.                                                                                                                          |
| Does OUT debit balance?                                   | No.                                                                                                                                                               |
| What happens on online authorization timeout?             | ONLINE_OFFLINE falls back to local cached rules; a valid server REJECT does not fall back.                                                                        |
| What marks a trip Synced?                                 | Successful image handling followed by MQTT QoS acknowledgement and local sync-state update.                                                                       |
| Where are operator vs server faults logged?               | Status Log for local/hardware/lane events; Server Log for upload/MQTT/sync events.                                                                                |
| Does storage retention delete the DB?                     | No. It targets dated image/log files.                                                                                                                             |
| <b>KT PASS CRITERIA</b>                                   | The engineer should be able to identify the failure layer (RFID, sensor/control, camera, local data, server sync or configuration) before taking recovery action. |

## Appendix A. Control Command Reference

| Purpose             | Command                  |
|---------------------|--------------------------|
| Open IN barrier     | OPEN IN BB               |
| Close IN barrier    | CLOSE IN BB              |
| Open OUT barrier    | OPEN OUT BB              |
| Close OUT barrier   | CLOSE OUT BB             |
| IN red              | IN RED [vehicle number]  |
| IN green            | IN GRN                   |
| IN orange           | IN ORG                   |
| IN buzzer ON        | IN Buzzer                |
| IN buzzer OFF       | IN Buzzer OFF            |
| IN all outputs off  | IN ALL OFF               |
| OUT red             | OUT RED [vehicle number] |
| OUT green           | OUT GRN                  |
| OUT orange          | OUT ORG                  |
| OUT buzzer ON       | OUT Buzzer               |
| OUT buzzer OFF      | OUT Buzzer OFF           |
| OUT all outputs off | OUT ALL OFF              |

### Sensor messages

| Lane | Detected     | Released                                                    |
|------|--------------|-------------------------------------------------------------|
| IN   | IN Detected  | IN Released (configured; common Released aliases accepted)  |
| OUT  | OUT Detected | OUT Released (configured; common Released aliases accepted) |

## Appendix B. Default Operational Values

| Area               | Default from Source                             | Commissioning Note                               |
|--------------------|-------------------------------------------------|--------------------------------------------------|
| IN RFID            | COM3, 115200, 8-N-1, UhfCfFrame                 | Verify actual Windows COM number.                |
| OUT RFID           | COM6, 115200, 8-N-1, UhfCfFrame                 | Verify actual Windows COM number.                |
| Wired Control      | COM10, 9600, LineText                           | Primary IND link.                                |
| Bluetooth Backup   | Enabled; COM11, 9600                            | Use actual paired SPP COM port.                  |
| Wi-Fi Backup       | Enabled; TCP endpoint configured in appsettings | Verify site IP/port before use.                  |
| Hardware reconnect | 5 seconds                                       | Reconnect loop enforces minimum practical delay. |
| Camera caching     | 1200 ms                                         | Tune only after stability test.                  |

| Area                          | Default from Source                  | Commissioning Note                                                                                                                                                                                                                        |
|-------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Camera watchdog               | 12 seconds                           | Frozen playback restart threshold.                                                                                                                                                                                                        |
| Camera reconnect              | 5 seconds                            | Per camera.                                                                                                                                                                                                                               |
| Server sync                   | 5 seconds                            | Pending trip cycle.                                                                                                                                                                                                                       |
| Sync batch                    | 25                                   | Maximum eligible items per cycle.                                                                                                                                                                                                         |
| Retry backoff                 | 15 to 300 seconds                    | Exponential retry.                                                                                                                                                                                                                        |
| Vehicle API auto sync         | 30 seconds                           | Separate from API request timeout.                                                                                                                                                                                                        |
| Storage retention             | 30 days images / 30 days logs        | 1-3650 days allowed in panel.                                                                                                                                                                                                             |
| Internet check                | 15 seconds; 4-second request timeout | Uses configured HTTP endpoints.                                                                                                                                                                                                           |
| Exceptional approval          | IN enabled / OUT enabled             | Protected setting.                                                                                                                                                                                                                        |
| Auto approval                 | IN disabled / OUT disabled           | Protected setting.                                                                                                                                                                                                                        |
| <b>VERIFY BEFORE SITE USE</b> |                                      | Defaults are source-package values, not a substitute for site commissioning. Confirm COM ports, reader mode, sensor strings, RTSP URLs, server endpoints, MQTT topics, image-upload protocol and security settings on the target machine. |



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## Appendix C. Production Readiness / Acceptance Checklist

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- [ ] Correct v1.0 standalone EXE deployed and version confirmed.
- [ ] Windows power/sleep settings are appropriate for continuous gate operation.
- [ ] Runtime configuration backed up and credentials access-restricted.
- [ ] IN RFID reader verified with real tags.
- [ ] OUT RFID reader verified with real tags.
- [ ] IN/OUT sensor Detected and Released messages verified.
- [ ] Wired IND control verified.
- [ ] Bluetooth failover verified if enabled.
- [ ] Wi-Fi control failover verified if enabled.
- [ ] Failback to wired verified without duplicate commands.
- [ ] IN/OUT barrier commands verified.
- [ ] RED/ORANGE/GREEN and Buzzer ON/OFF verified on both lanes.
- [ ] Cross-lane same-RFID collision guard verified.
- [ ] IN and OUT RTSP streams stable.
- [ ] Snapshot filename/path verified.
- [ ] FREE IN/OUT test completed.
- [ ] PAID normal balance test completed.
- [ ] One-negative-trip rule test completed in offline fallback.
- [ ] Contractor balance path tested when used.
- [ ] Missing OUT manual/auto approval tested as configured.
- [ ] Missing IN manual/auto approval tested as configured.
- [ ] Vehicle import and API auto sync tested.
- [ ] Server MQTT publish and Pending -> Synced transition verified.
- [ ] SFTP/FTP/FTPS image upload mode verified.
- [ ] FTPS certificate validation/pinning verified if applicable.
- [ ] Recharge command, duplicate recharge and acknowledgement tested.
- [ ] Network/server outage test confirms local gate operation continues.
- [ ] Storage retention settings reviewed.
- [ ] Backup/restore procedure tested on a non-production copy.
- [ ] Operator/support KT exercises completed.

Prepared from: RealTech\_iTOLL\_Automation\_v1.0 (Final Production) source package.

## Appendix D. MQTT JSON Data Packet Samples

This appendix lists the MQTT JSON message types used by the final production application. All IDs, balances, vehicle numbers, names and timestamps below are sample values only. No live broker, camera, FTP/SFTP or password credentials are included.

|                     |                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VERSION NOTE</b> | The final production code emits gate transaction schemaVersion 2.0 and online authorization request schemaVersion 2.0. Recharge command/acknowledgement remains schemaVersion 1.0. Older interface documents in the source package may show earlier gate schema examples. |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### D.1 MQTT Topic / Packet Map

| Topic                                    | Direction        | Packet                                                                                                                                      | QoS / Retain                                                               |
|------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| IWS_STP-001/Device_Response              | Device -> Server | Physical gate transaction (IN or OUT)                                                                                                       | QoS 1 / false by default                                                   |
| IWS_STP-001/Trip_Authorization/Request   | Device -> Server | Online trip authorization request                                                                                                           | Configured publish QoS (default 1) / false                                 |
| IWS_STP-001/Trip_Authorization/Response  | Server -> Device | Online trip authorization response                                                                                                          | Server should publish QoS 0 for current response parser                    |
| IWS_STP-001/STP_RECHARGE_FROM_SERVER     | Server -> Device | RFID recharge command                                                                                                                       | Subscribe QoS follows configured QoS (default 1); retain false recommended |
| IWS_STP-001/STP_RECHARGE_ACK_FROM_DEVICE | Device -> Server | RFID recharge application acknowledgement                                                                                                   | QoS 1 / false by default                                                   |
| <b>MQTTX</b>                             |                  | For device monitoring, IWS_STP-001/# covers the device-root topics. Use a unique MQTT client ID. Do not publish retained recharge commands. |                                                                            |

### D.2 Normal IN Gate Transaction

**Topic:** IWS\_STP-001/Device\_Response **Direction:** Device -> Server **QoS:** 1 **Retain:** false

```
{
  "schemaVersion": "2.0",
  "transactionId": "8d1d31e9-9a38-472c-a3a0-df88f5bc52e7",
  "siteId": "SITE-001",
  "deviceId": "IWS_STP-001",
  "laneId": "MAIN-GATE",
  "deviceName": "Tambaram iTOLL 001",
  "direction": "IN",
  "rfidNumber": "E20034120123456789012345",
  "vehicleNumber": "TN11AB1234",
  "vehicleCategory": "10000",
  "accessType": "Paid",
  "contractorId": "",
  "contractorName": "",
  "balanceType": "RFID",
  "authorizationSource": "OFFLINE_LOCAL",
  "authorizationRequestId": "a41088a4fa134c4a96ec667f740ebc09",
  "serverDecision": "UNAVAILABLE_FALLBACK",
  "serverReason": "Online authorization disabled",
  "previousBalance": 500.0,
  "debitAmount": 50.0,
  "newBalance": 450.0,
  "imageCaptureStatus": "Captured",
  "imageFileName": "iTOLL-2026-09-11-141500-IN.png",
  "imageRemotePath": "/vehicle-images/iTOLL-2026-09-11-141500-IN.png",
  "tripDate": "2026-09-11",
  "tripTime": "14:15:00.125",
  "timeZoneOffset": "+05:30",
}
```

```
{
  "processedAt": "2026-09-11T14:15:00.125+05:30",
  "notes": "Processed locally; server upload is asynchronous."
}
```

### D.3 Normal OUT Gate Transaction

**Topic:** IWS\_STP-001/Device\_Response **Direction:** Device -> Server **QoS:** 1 **Retain:** false

```
{
  "schemaVersion": "2.0",
  "transactionId": "0f109d67-d93e-4ee0-9d55-39b37ca7a1c4",
  "siteId": "SITE-001",
  "deviceId": "IWS_STP-001",
  "laneId": "MAIN-GATE",
  "deviceName": "Tambaram iTOLL 001",
  "direction": "OUT",
  "rfidNumber": "E20034120123456789012345",
  "vehicleNumber": "TN11AB1234",
  "vehicleCategory": "10000",
  "accessType": "Paid",
  "contractorId": "",
  "contractorName": "",
  "balanceType": "RFID",
  "authorizationSource": "OFFLINE_LOCAL",
  "authorizationRequestId": "1c85af89f08f460dbedff34fd309867e",
  "serverDecision": "UNAVAILABLE_FALLBACK",
  "serverReason": "Online authorization disabled",
  "entryTransactionId": "8d1d31e9-9a38-472c-a3a0-df88f5bc52e7",
  "previousBalance": 450.0,
  "debitAmount": 0.0,
  "newBalance": 450.0,
  "imageCaptureStatus": "Captured",
  "imageFileName": "iTOLL-2026-09-11-142000-OUT.png",
  "imageRemotePath": "/vehicle-images/iTOLL-2026-09-11-142000-OUT.png",
  "tripDate": "2026-09-11",
  "tripTime": "14:20:00.410",
  "timeZoneOffset": "+05:30",
  "processedAt": "2026-09-11T14:20:00.410+05:30",
  "notes": "Processed locally; server upload is asynchronous."
}
```

### D.4 OUT Transaction with Missing IN Approval

**Topic:** IWS\_STP-001/Device\_Response **Direction:** Device -> Server **QoS:** 1 **Retain:** false

```
{
  "schemaVersion": "2.0",
  "transactionId": "9f60cb1a-faf4-45fd-8840-b94ae5ac9f2c",
  "siteId": "SITE-001",
  "deviceId": "IWS_STP-001",
  "laneId": "MAIN-GATE",
  "deviceName": "Tambaram iTOLL 001",
  "direction": "OUT",
  "rfidNumber": "E20034120123456789012345",
  "vehicleNumber": "TN11AB1234",
  "vehicleCategory": "10000",
  "accessType": "Paid",
  "contractorId": "",
  "contractorName": "",
  "balanceType": "RFID",
  "authorizationSource": "OFFLINE_LOCAL",
  "authorizationRequestId": "a8b673e6a54649169be7aed87bc4a882",
  "serverDecision": "UNAVAILABLE_FALLBACK",
  "serverReason": "Online authorization disabled",
  "previousBalance": 450.0,
  "debitAmount": 0.0,
  "newBalance": 450.0,
  "imageCaptureStatus": "Captured",
  "imageFileName": "iTOLL-2026-09-11-143000-OUT.png",
  "imageRemotePath": "/vehicle-images/iTOLL-2026-09-11-143000-OUT.png",
  "tripDate": "2026-09-11",
  "tripTime": "14:30:00.500",
  "timeZoneOffset": "+05:30",
  "processedAt": "2026-09-11T14:30:00.500+05:30",
}
```

```

"exceptionalApproval": {
  "approvalMode": "Manual",
  "reason": "Missing IN Trip",
  "approverName": "Shift Supervisor",
  "approverRole": "Supervisor",
  "approverMobile": "9876543210",
  "approvalDate": "2026-09-11",
  "approvalTime": "14:29:58.300",
  "timeZoneOffset": "+05:30"
},
"missingTripNotification": {
  "missingDirection": "IN",
  "notificationOnly": true,
  "syntheticTripCreated": false,
  "countedAsTransaction": false,
  "message": "Expected IN trip was missing. Notification only; no separate missing trip was created, uploaded, or counted."
},
"notes": "Processed locally; server upload is asynchronous. Exceptional approval: Missing IN Trip."
}

```

- entryTransactionId is omitted because no real matching IN trip exists.

## D.5 IN Transaction with Missing OUT Approval

**Topic:** IWS\_STP-001/Device\_Response **Direction:** Device -> Server **QoS:** 1 **Retain:** false

```

{
  "schemaVersion": "2.0",
  "transactionId": "7718a2b2-622c-44c8-985e-9bac3b75038e",
  "siteId": "SITE-001",
  "deviceId": "IWS_STP-001",
  "laneId": "MAIN-GATE",
  "deviceName": "Tambaram iTOLL 001",
  "direction": "IN",
  "rfidNumber": "E20034120123456789012345",
  "vehicleNumber": "TN11AB1234",
  "vehicleCategory": "10000",
  "accessType": "Paid",
  "contractorId": "",
  "contractorName": "",
  "balanceType": "RFID",
  "authorizationSource": "OFFLINE_LOCAL",
  "authorizationRequestId": "e5b9241320e6401baaab2a59198dff9e",
  "serverDecision": "UNAVAILABLE_FALLBACK",
  "serverReason": "Online authorization disabled",
  "previousBalance": 450.0,
  "debitAmount": 50.0,
  "newBalance": 400.0,
  "imageCaptureStatus": "Captured",
  "imageFileName": "iTOLL-2026-09-11-144000-IN.png",
  "imageRemotePath": "/vehicle-images/iTOLL-2026-09-11-144000-IN.png",
  "tripDate": "2026-09-11",
  "tripTime": "14:40:00.600",
  "timeZoneOffset": "+05:30",
  "processedAt": "2026-09-11T14:40:00.600+05:30",
  "exceptionalApproval": {
    "approvalMode": "Auto",
    "reason": "Missing OUT Trip",
    "approverName": "SYSTEM",
    "approverRole": "Auto Approval",
    "approverMobile": "",
    "approvalDate": "2026-09-11",
    "approvalTime": "14:39:58.700",
    "timeZoneOffset": "+05:30"
  },
  "missingTripNotification": {
    "missingDirection": "OUT",
    "notificationOnly": true,
    "syntheticTripCreated": false,
    "countedAsTransaction": false,
    "message": "Expected OUT trip was missing. Notification only; no separate missing trip was created, uploaded, or counted."
  },
  "notes": "Processed locally; server upload is asynchronous. Exceptional approval: Missing OUT Trip."
}

```

- The earlier unmatched IN is closed internally. Only this new physical IN transaction is published; no synthetic OUT packet is created.

## D.6 Online Trip Authorization Request

**Topic:** IWS\_STP-001/Trip\_Authorization/Request **Direction:** Device -> Server **QoS:** configured (default 1) **Retain:** false

```
{
  "schemaVersion": "2.0",
  "requestId": "48d8cdb8dbd94ee18c0b1ce82a93aa51",
  "siteId": "SITE-001",
  "deviceId": "IWS_STP-001",
  "direction": "IN",
  "vehicleNumber": "TN11AB1234",
  "rfidNumber": "E20034120123456789012345",
  "vehicleCategory": "10000",
  "categoryDebitAmount": 50.0,
  "contractorId": "",
  "contractorName": "",
  "balanceType": "RFID",
  "deviceLocalBalance": 500.0,
  "deviceDateTime": "2026-09-11T14:45:00.100+05:30",
  "device": {
    "siteId": "SITE-001",
    "deviceId": "IWS_STP-001",
    "machineName": "GATE-PC-01"
  }
}
```

## D.7 Online Trip Authorization Response - APPROVED

**Topic:** IWS\_STP-001/Trip\_Authorization/Response **Direction:** Server -> Device **QoS:** 0 recommended/required by current parser **Retain:** false

```
{
  "requestId": "48d8cdb8dbd94ee18c0b1ce82a93aa51",
  "status": "APPROVED",
  "approved": true,
  "balance": 500.0,
  "debitAmount": 50.0,
  "reason": "Vehicle authorized by server"
}
```

- requestId must exactly match the request. The device treats approved=true OR status=APPROVED as approval.

## D.8 Online Trip Authorization Response - REJECTED

**Topic:** IWS\_STP-001/Trip\_Authorization/Response **Direction:** Server -> Device **QoS:** 0 recommended/required by current parser **Retain:** false

```
{
  "requestId": "48d8cdb8dbd94ee18c0b1ce82a93aa51",
  "status": "REJECTED",
  "approved": false,
  "balance": -25.0,
  "debitAmount": 50.0,
  "reason": "Insufficient server balance"
}
```

- A valid server rejection is final. Offline fallback is used only when the online authorization is unavailable or times out.

## D.9 RFID Recharge Command

**Topic:** IWS\_STP-001/STP\_RECHARGE\_FROM\_SERVER **Direction:** Server -> Device **QoS:** 1 **Retain:** false

```
{
  "schemaVersion": "1.0",
  "messageType": "rfidRecharge",
  "rechargeId": "RCH-20260911-000001",
  "siteId": "SITE-001",
}
```

```
{
  "deviceId": "IWS_STP-001",
  "rfidNumber": "E20034120123456789012345",
  "vehicleNumber": "TN11AB1234",
  "previousBalance": -50.0,
  "rechargeAmount": 500.0,
  "newBalance": 450.0,
  "rechargedAt": "2026-09-11T14:50:00+05:30",
  "paymentReference": "PAY-20260911-77881",
  "operatorId": "OP-1007",
  "operatorName": "Recharge Operator",
  "remarks": "Cash recharge at site office"
}
```

- rechargeId is the idempotency key. A retry must reuse the same rechargeId; a new ID means a new credit operation.

## D.10 RFID Recharge Acknowledgement - Applied

**Topic:** IWS\_STP-001/STP\_RECHARGE\_ACK\_FROM\_DEVICE **Direction:** Device -> Server **QoS:** 1 **Retain:** false

```
{
  "schemaVersion": "1.0",
  "messageType": "rfidRechargeAck",
  "rechargeId": "RCH-20260911-000001",
  "siteId": "SITE-001",
  "deviceId": "IWS_STP-001",
  "laneId": "MAIN-GATE",
  "deviceName": "Tambaram iTOLL 001",
  "rfidNumber": "E20034120123456789012345",
  "vehicleNumber": "TN11AB1234",
  "status": "Applied",
  "success": true,
  "duplicate": false,
  "rechargeAmount": 500.0,
  "previousBalance": -50.0,
  "newBalance": 450.0,
  "serverExpectedNewBalance": 450.0,
  "serverBalanceMatched": true,
  "processedAt": "2026-09-11T14:50:01.240+05:30",
  "message": "Recharge applied to the local RFID balance."
}
```

## D.11 RFID Recharge Acknowledgement - Duplicate

**Topic:** IWS\_STP-001/STP\_RECHARGE\_ACK\_FROM\_DEVICE **Direction:** Device -> Server **QoS:** 1 **Retain:** false

```
{
  "schemaVersion": "1.0",
  "messageType": "rfidRechargeAck",
  "rechargeId": "RCH-20260911-000001",
  "siteId": "SITE-001",
  "deviceId": "IWS_STP-001",
  "laneId": "MAIN-GATE",
  "deviceName": "Tambaram iTOLL 001",
  "rfidNumber": "E20034120123456789012345",
  "vehicleNumber": "TN11AB1234",
  "status": "Duplicate",
  "success": true,
  "duplicate": true,
  "rechargeAmount": 500.0,
  "previousBalance": -50.0,
  "newBalance": 450.0,
  "serverExpectedNewBalance": 450.0,
  "serverBalanceMatched": true,
  "processedAt": "2026-09-11T14:50:01.240+05:30",
  "message": "Recharge ID was already processed with status Applied: Recharge applied to the local RFID balance."
}
```

## D.12 RFID Recharge Acknowledgement - Rejected

**Topic:** IWS\_STP-001/STP\_RECHARGE\_ACK\_FROM\_DEVICE **Direction:** Device -> Server **QoS:** 1 **Retain:** false

```
{
  "schemaVersion": "1.0",
  "messageType": "rfidRechargeAck",
  "rechargeId": "RCH-20260911-000002",
  "siteId": "SITE-001",
  "deviceId": "TWS_STP-001",
  "laneId": "MAIN-GATE",
  "deviceName": "Tambaram iTOLL 001",
  "rfidNumber": "E20034120999999999999999",
  "vehicleNumber": "",
  "status": "Rejected",
  "success": false,
  "duplicate": false,
  "rechargeAmount": 500.0,
  "serverExpectedNewBalance": 500.0,
  "processedAt": "2026-09-11T14:51:03.410+05:30",
  "message": "RFID is not registered."
}
```

## D.13 MQTT Integration Rules to Remember

- Gate transaction synchronization uses the physical transactionId as the unique device transaction ID.
- Normal gate packets omit exceptionalApproval and missingTripNotification because the serializer omits null properties.
- For missing-trip approval, only the current physical IN or OUT packet is published. No second/synthetic missing-direction MQTT transaction is sent.
- Online authorization response requestId must match the requestId. A mismatched response is treated as unavailable and local fallback may be used.
- Current authorization response parsing expects the server PUBLISH packet without a QoS packet identifier; publish the response at QoS 0.
- Recharge commands should not be retained. Use QoS 1 and application acknowledgement/retry with the same rechargeId.
- Recharge acknowledgements may be Applied, Duplicate or Rejected. Duplicate protects against double credit.
- Gate Device\_Response uses broker PUBACK as synchronization success; there is no separate gate application-level ACK packet in the current production implementation.

MQTT packet appendix verified against final production source: ServerSyncWorker, OnlineTripAuthorizationService, MqttRechargeSyncWorker, RfidRecharge models and appsettings.json.